

RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

FDN 1204/ FDN 1306 Basic Mathematics
Tutorial 1

1. In a survey of university students, 64 had taken mathematics course, 94 had taken chemistry course, 58 had taken physics course, 28 had taken mathematics and physics, 26 had taken mathematics and chemistry, 22 had taken chemistry and physics course, and 14 had taken all the three courses. Find how many had taken one course only.
2. In a group of students, 65 play football, 45 play hockey, 42 play cricket, 20 play foot ball and hockey, 25 play football and cricket, 15 play hockey and cricket and 8 play all the three games. Find the total number of students in the group (Assume that each student in the group plays at least one game).
3. In a town 85% of the people speak Tamil, 40% speak English and 20% speak Hindi. Also 32% speak Tamil and English, 13% speak Tamil and Hindi and 10% speak English and Hindi, find the percentage of people who can speak all the three languages.
4. An advertising agency finds that, of its 170 clients, 115 use Television, 110 use Radio and 130 use Magazines. Also 85 use Television and Magazines, 75 use Television and Radio, 95 use Radio and Magazines, 70 use all the three. Draw Venn diagram to represent these data. Find
 - a) how many use only Radio?
 - b) how many use only Television?
 - c) how many use Television and Magazine but not radio?
5. A survey of faculty and graduate students at the University of Florida's film school revealed the following information: 51 admire Moe 49 admire Larry 60 admire Curly 34 admire Moe and Larry 32 admire Larry and Curly 36 admire Moe and Curly 24 admire all three of the Stooges 1 admires none of the Three Stooges
 - a) How many people were surveyed?
 - b) How many admire Curly, but not Larry nor Moe?

c) How many admire Larry or Curly?

d) How many admire exactly one of the Stooges? e) How many admire exactly two of the Stooges?

6. Fifty elephants were asked about things that they like. 8 like tigers 6 like ivory carvings 9 like mice 1 likes all three of these things 4 like tigers and ivory carvings 5 like tigers and mice 3 like mice and ivory carvings A. How many like none of these things? B. How like ivory carvings but don't like tigers? C. How many like fewer than two of these things? D. How many like mice or ivory carvings?
7. The members of the Gainesville Film Critics Society were surveyed regarding their cinematic heroes. 38 admire Moe 36 admire Larry 21 admire Curly 23 admire Moe and admire Larry 5 admire Larry and admire Curly 7 admire Moe and admire Curly 3 admire Moe and admire Larry and admire Curly 16 don't admire Moe and don't admire Larry and don't admire Curly A. How many were surveyed? B. How many admire at least one of the Stooges? C. How many admire Moe or Larry?
8. A number of Yugo owners were surveyed regarding their favorite activities. 57 enjoy walking 65 enjoy waiting for tow trucks 60 enjoy hitchhiking 20 enjoy waiting for tow trucks and enjoy hitchhiking and enjoy walking 22 enjoy hitchhiking and enjoy walking 12 don't enjoy waiting for tow trucks and don't enjoy hitchhiking and don't enjoy walking 23 enjoy walking but don't enjoy hitchhiking and don't enjoy waiting for tow trucks 38 enjoy waiting for tow trucks and enjoy hitchhiking A. How many were surveyed? B. How many enjoy at least two of these activities?
9. 100 Neanderthals were surveyed regarding their favorite possessions. 13 own a bone and own a stick 32 own a stick 3 own a rock and own a bone and own a stick 30 own a rock 14 own a rock and own a bone 31 own a bone 12 own a rock and own a stick How many own none of these things?
10. In a survey of 130 people, the following data were collected: 106 people subscribed to the newspaper, 29 people subscribed to magazines, and 17 people were members of a mail CD club. Seventeen subscribed to both the newspaper and the magazines, 5 people subscribed to magazines and were members of a CD club, and 10 people subscribed to the newspaper and were members of a mail CD club. Three people subscribed to both the newspaper and magazines and were members of a mail CD club. Make and fill in a Venn diagram to illustrate this situation.